

# Aircraft Price Analysis

Statistical Learning project

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# Introduction and Key Questions

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# Objective of the Analysis

Predict the price of aircraft using technical features and linear regression models with regularization.

# Key Questions About the Dataset

- qui mettiamo 2 domande che ci facciamo sul dataset  
xxxxxxxxxxxxdsuviuwiofhwiohvioshiohvoiqervhoqeirhoeqrig.
- Q: What is thegrsgjoiejgroiegjoqregtarget and in what units?

# Dataset Description

- **Total Observations:** 507
- **Total Variables:** 14
- **Numerical Variables:** 12
- **Categorical Variables:** 2
- **Target:** price (price in USD)
- **Source:** Kaggle – Aircraft Price Analysis & Prediction Dataset

# Numerical Variables 1

- `engine_power`: engine power (HP)
- `max_speed`: maximum speed (km/h)
- `cruise_speed`: cruise speed (km/h)
- `stall_speed`: stall speed (km/h)
- `fuel_tank`: fuel tank capacity (liters)
- `range`: range (km)
- `length`: aircraft length (m)

## Numerical Variables 2

- `wing_span`: wingspan (m)
- `empty_weight`: empty weight (kg)
- `all_eng_roc`: rate of climb with all engines (ft/min)
- `one_eng_roc`: rate of climb with one engine out (ft/min)
- `takeoff_distance`: takeoff distance (m)
- `landing_distance`: landing distance (m)

# Categorical Variables

- `model_name`: aircraft model name
- `engine_type`: type of engine



# Data Preprocessing

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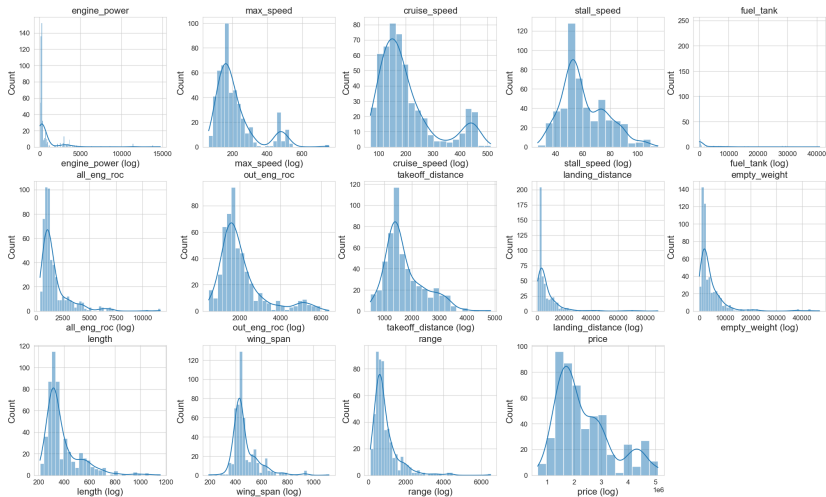
- Removal of rows with missing values in the target price
- Checking and handling of outliers
- Necessary transformations to normalize distributions

# Missing Data

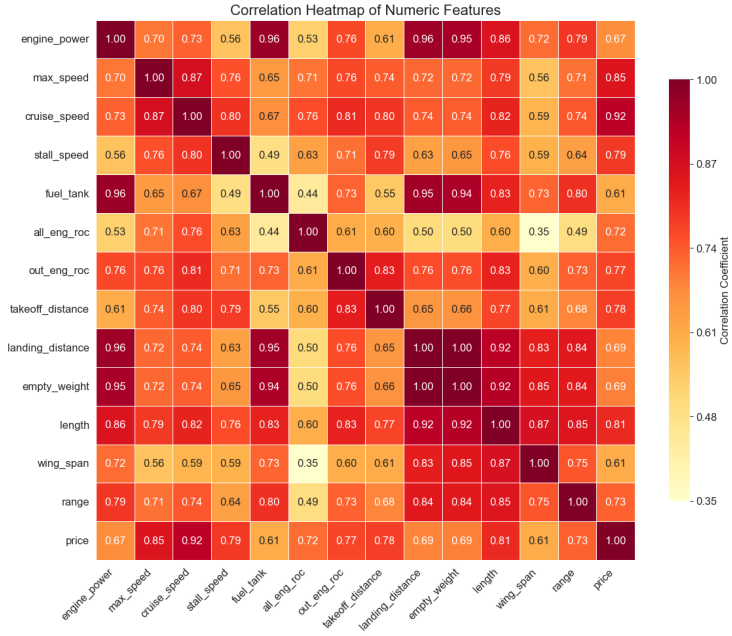
- 10 rows removed due to missing price
- Final dataset: 497 observations



# Feature Distributions



# Correlation Matrix



- Numerical variables exhibit skewed distributions
- Correlations between variables suggest multicollinearity
- Transformations needed to improve linearity and normality

# Log Transformation

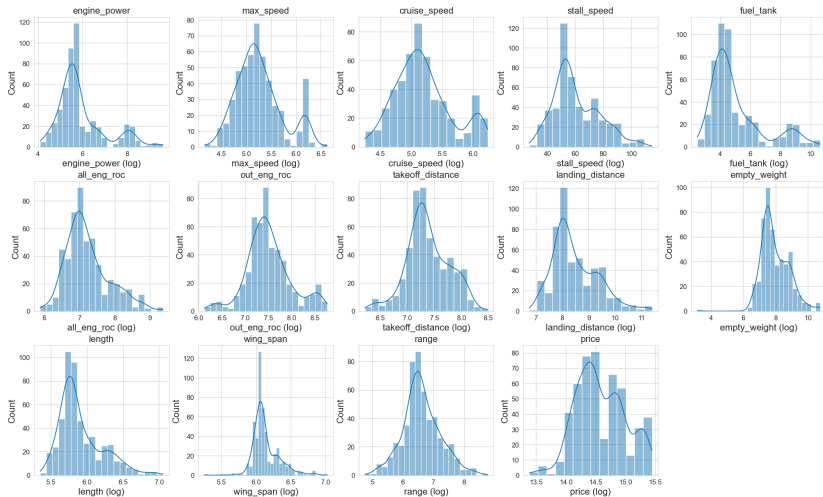
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# Motivations for Log Transformation

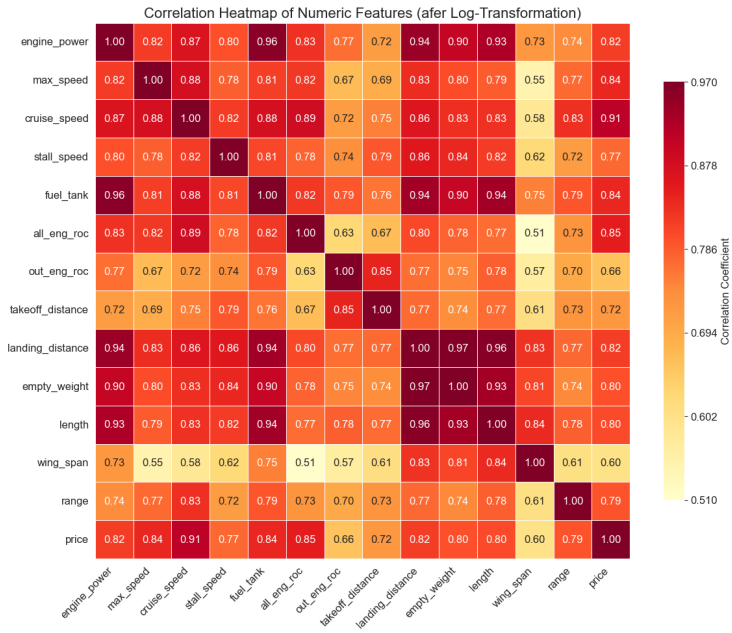
- Numerical variables have **skewed** distributions
- Logarithmic transformation applied to all numerical features to:
  - Achieve more symmetric distributions
  - Improve linearity between features
  - Reduce the impact of outliers
  - Stabilize variance



# Distributions After Log Transformation



# Correlation Matrix After Log Transformation



# Linear Regression

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# Multiple Linear Regression Model

$$\log(\text{price}) = \beta_0 + \sum_{j=1}^{13} \beta_j \log(x_j + 1) + \varepsilon$$

- Preprocessing: log-transform of all numerical variables
- Validation: 10-fold CV with shuffling, `random_state=42`

## K-Fold CV Metrics (Full)

=== K-Fold CV Metrics ===

$R^2$  mean = 0.8282  $\pm$  0.0585

MSE mean = 0.0275  $\pm$  0.0080

RMSE (de-logged) = 391503.27  $\pm$  0.0080

Inferential summary aggregated over the 10 training folds is shown in the next slide.

## Inferential Summary (Full)

| Feature          | coef <sub>mean</sub> | coef <sub>std</sub> | t <sub>mean</sub> | p <sub>mean</sub> |
|------------------|----------------------|---------------------|-------------------|-------------------|
| engine_power     | 0.011                | 0.021               | 0.37              | 0.543             |
| max_speed        | 0.175                | 0.026               | 4.64              | 0.001             |
| cruise_speed     | 0.504                | 0.039               | 9.73              | 0.001             |
| stall_speed      | 0.001                | 0.001               | 0.87              | 0.430             |
| fuel_tank        | 0.039                | 0.007               | 2.19              | 0.038             |
| all_eng_roc      | 0.119                | 0.016               | 4.18              | 0.001             |
| out_eng_roc      | -0.099               | 0.006               | -2.90             | 0.005             |
| takeoff_distance | 0.121                | 0.009               | 2.91              | 0.005             |
| landing_distance | -0.325               | 0.059               | -5.05             | 0.001             |
| empty_weight     | 0.104                | 0.050               | 2.66              | 0.008             |
| length           | 0.155                | 0.046               | 1.48              | 0.178             |
| wing_span        | 0.367                | 0.078               | 3.88              | 0.001             |
| range            | 0.025                | 0.011               | 1.06              | 0.336             |

## Reduced Model (Only Significant Variables)

- Selected the 9 features with  $p < 0.05$
- **Removed Features:** engine\_power, stall\_speed, length, range
- Retrained with the same CV settings

## Reduced Model vs Full Model Results

| Full Model          |                     | Reduced Model       |                     |
|---------------------|---------------------|---------------------|---------------------|
| (14 features)       |                     | (9 features)        |                     |
| R <sup>2</sup> mean | 0.8282 $\pm$ 0.0585 | R <sup>2</sup> mean | 0.8291 $\pm$ 0.0643 |
| MSE mean            | 0.0275 $\pm$ 0.0080 | MSE mean            | 0.0271 $\pm$ 0.0080 |
| RMSE (USD)          | \$391,503.27        | RMSE (USD)          | \$389,179.25        |

- Slight improvement in R<sup>2</sup> and RMSE

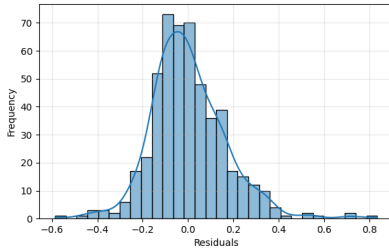


## Coefficients of Reduced Model

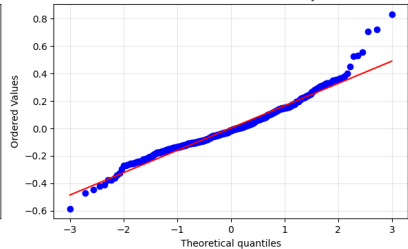
| Feature          | coef <sub>mean</sub> | coef <sub>std</sub> | t <sub>mean</sub> | p <sub>mean</sub> |
|------------------|----------------------|---------------------|-------------------|-------------------|
| intercept        | 8.515                | 0.311               | 16.37             | 0.001             |
| max_speed        | 0.179                | 0.024               | 4.84              | 0.001             |
| cruise_speed     | 0.522                | 0.032               | 10.81             | 0.001             |
| fuel_tank        | 0.049                | 0.005               | 3.60              | 0.001             |
| all_eng_roc      | 0.125                | 0.014               | 4.50              | 0.001             |
| out_eng_roc      | -0.084               | 0.007               | -2.57             | 0.013             |
| takeoff_distance | 0.132                | 0.006               | 3.44              | 0.001             |
| landing_distance | -0.285               | 0.059               | -5.43             | 0.001             |
| empty_weight     | 0.109                | 0.057               | 2.82              | 0.006             |
| wing_span        | 0.405                | 0.057               | 4.75              | 0.001             |

# Residual Analysis

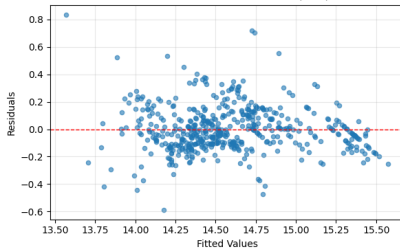
Residuals Distribution (Test Sets)



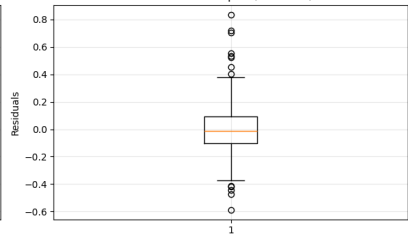
Q-Q Plot - Residuals Normality



Residuals vs Predicted Values (Test)



Residuals Boxplot (Test Sets)



## Residual Analysis: Observations

- Residuals exhibit a symmetric distribution
- No obvious patterns, suggesting the model captures the variable relationships well
- A few observations have larger residuals, but no extreme outliers

# Regularization

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# Ridge Regression

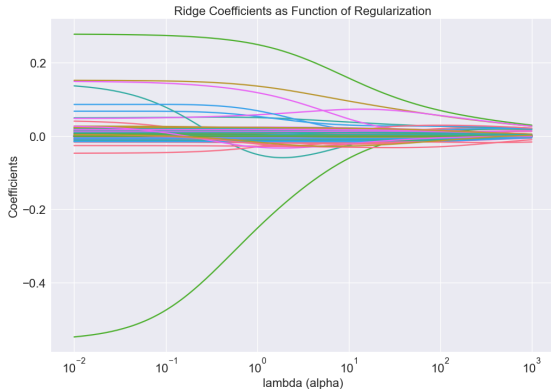
$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2 \right\}$$

- L2 penalty on coefficients
- Uses all features
- Cross-validation to select the best  $\alpha$
- Objective: reduce overfitting and improve generalization

# Ridge: Metrics and Coefficients

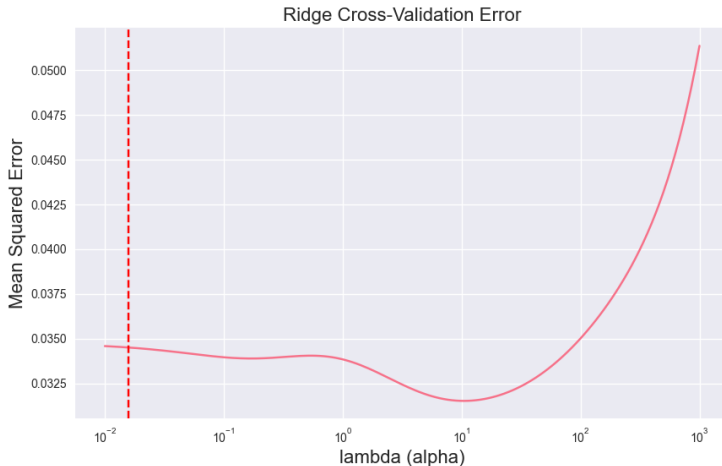
## Best $\alpha$ and Metrics

| Parameter     | Value   |
|---------------|---------|
| Best $\alpha$ | 13.2070 |
| Test MSE      | 0.0249  |
| Test RMSE     | 0.1579  |
| Test MAE      | 0.1182  |
| Test $R^2$    | 0.8584  |



Ridge coefficients after regularization.

## Ridge: Cross-Validation Error



Cross-validation error as a function of  $\alpha$  for selecting the optimal parameter.

$$\min_{\beta} \left\{ \frac{1}{2n} \|y - X\beta\|_2^2 + \alpha \|\beta\|_1 \right\}$$

- L1 penalty: automatic feature selection
- Coefficients can be driven to zero
- Cross-validation to select the best  $\alpha$
- Objective: interpretability and dimensionality reduction



# Lasso: Metrics and Coefficients

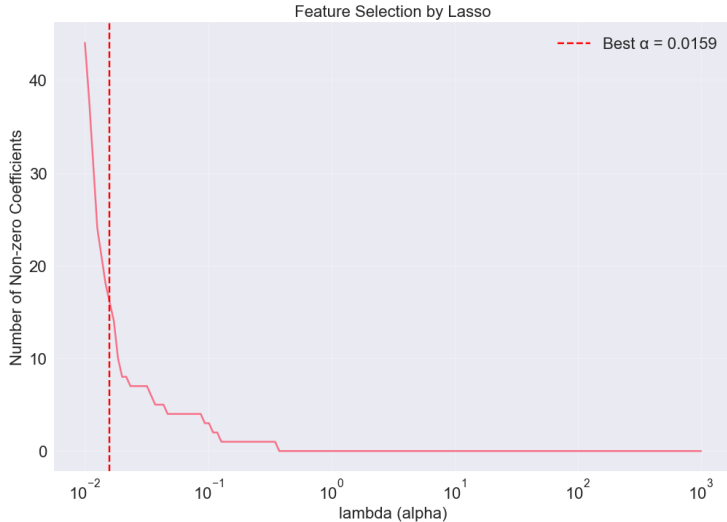
## Best $\alpha$ and Metrics

| Parameter         | Value    |
|-------------------|----------|
| Best $\alpha$     | 0.0159   |
| Test MSE          | 0.0257   |
| Test RMSE         | 0.1604   |
| Test MAE          | 0.1192   |
| Test $R^2$        | 0.8539   |
| Selected Features | 16 / 290 |



Lasso coefficients after regularization.

# Lasso: Feature Selection



*[Graph not available: LassoCV.png]*

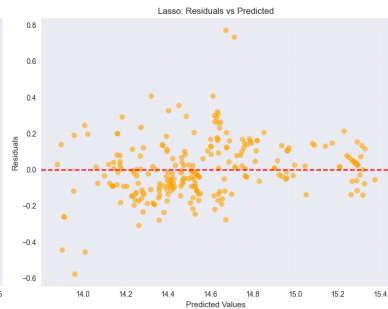
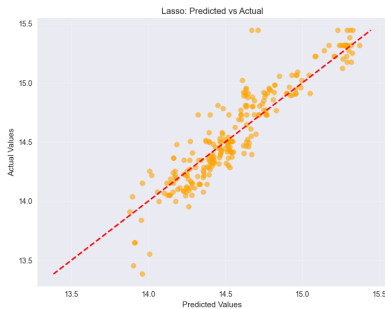
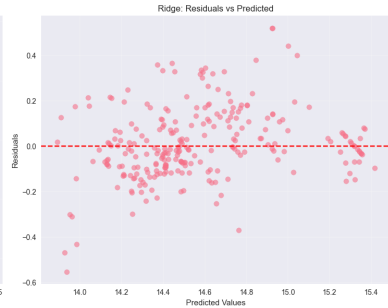
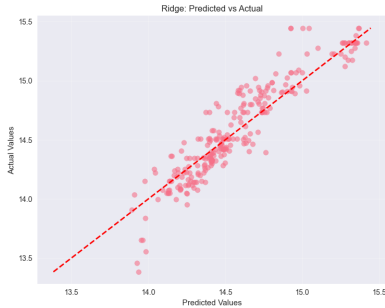
## Final Comparison

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## Ridge vs Lasso Comparison

| Metric   | Ridge   | Lasso  | Difference |
|----------|---------|--------|------------|
| MSE      | 0.0249  | 0.0257 | -0.0008    |
| RMSE     | 0.1579  | 0.1604 | -0.0025    |
| MAE      | 0.1182  | 0.1192 | -0.0010    |
| $R^2$    | 0.8584  | 0.8539 | 0.0045     |
| Features | 290     | 16     | 274        |
| $\alpha$ | 13.2070 | 0.0159 | —          |

# Residual Analysis



## Residual Analysis: Observations

- Ridge and Lasso residuals show symmetry and approximate normality
- No clear patterns, suggesting both models capture relationships well
- Ridge has slightly smaller residuals, indicating better fit
- Lasso remains interpretable with fewer features and similar performance

# Comparison of Linear Models

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# Comparison of Linear Models

| Model             | $R^2$  | RMSE (log) | RMSE (USD) |
|-------------------|--------|------------|------------|
| Linear Regression | 0.8291 | 0.1646     | \$389,179  |
| Ridge             | 0.8584 | 0.1579     | \$370,000  |
| Lasso             | 0.8539 | 0.1604     | \$376,000  |

## Observations

- Ridge achieves the best  $R^2$  (0.8584) and lowest RMSE
- Lasso remains competitive with only 16 features
- Regularization improves performance over simple linear models
- Minimal differences between Ridge and Lasso suggest robustness



# Generalized Additive Models (GAM)

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## Generalized Additive Models (GAM)

GAMs extend linear models by allowing non-linear relationships:

$$y = \beta_0 + f_1(x_1) + f_2(x_2) + \dots + f_p(x_p) + \epsilon \quad (1)$$

where  $f_i$  are smooth functions (splines) of the predictor variables.

- **Flexibility:** Capture complex non-linear relationships
- **Interpretability:** Partial effects visualizable for each feature
- **Regularization:** Control smoothness via  $\lambda$  parameters
- **Additivity:** Retain interpretable structure similar to linear models

## Model Setup

- **Target:** Log-price (consistent with Ridge/Lasso)
- **Features:** range, roc\_mean, speed\_margin, power\_per\_distance, wing\_span
- **Spline terms:**  $s()$  for continuous,  $f()$  for categorical
- **Optimization:** Automated grid-search for  $\lambda$

## Fitting Process

1. Cross-validation to select optimal  $\lambda$  for each term
2. Smoothing splines with automatic degrees-of-freedom control
3. Validation via AIC and deviance explained

## GAM Performance: Key Results

| Model    | Pseudo-R <sup>2</sup> | RMSE (log) | RMSE (USD) |
|----------|-----------------------|------------|------------|
| GAM Full | 0.892                 | 0.134      | \$315,000  |
| Ridge    | 0.858                 | 0.158      | \$370,000  |
| Lasso    | 0.854                 | 0.160      | \$376,000  |
| Linear   | 0.829                 | 0.165      | \$389,179  |

### Significant Improvements

- +3.4% Pseudo-R<sup>2</sup> compared to Ridge
- -15.2% RMSE compared to Ridge (\$55,000 less)
- -19.0% RMSE compared to linear regression
- AIC significantly improved

# Partial Dependence Plots: Interpretability

## Key Non-Linear Insights

- **Range:** Non-linear relationship with plateau beyond 4000 km
- **ROC\_mean:** Logarithmic growth with diminishing returns
- **Power\_per\_distance:** Quadratic relationship with local optimum
- **Wing\_span:** Threshold effect with breakpoint at 15 m
- **Speed\_margin:** Distinct categorical effects for different classes

## Interpretation of Partial Dependence

The plots show the isolated effect of each feature on the log-price, holding all else constant. This allows:

- Identification of critical thresholds
- Detection of complex non-linear patterns

## Model Validation

The residual analysis confirms the superior quality of the GAM:

- **Residuals well-distributed** around zero
- **Homoscedasticity** (no pattern in fitted values)
- **Normal distribution** of residuals
- **Better fit** than linear models

## Comparison with Linear Models

- Residual standard deviation: 0.134 (vs 0.158 Ridge)
- 15% reduction in unexplained variability
- No systematic patterns in residuals

# GAM vs Linear Models: Direct Comparison

## GAM Advantages

- Superior performance (-15% RMSE)
- Captures real non-linearities
- Maintains interpretability
- Better residuals
- Identification of thresholds

## Trade-offs

- Higher computational complexity
- Risk of overfitting
- More parameters to optimize
- Requires careful validation

## Recommendation

GAMs offer the best trade-off between performance and interpretability for this dataset, justifying the added complexity with significant prediction improvements.

## Conclusions and Next Steps

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## Next Steps

- Explore other non-linear models (Random Forest, SVM)
- Ensemble methods combining GAM with linear models
- Feature engineering based on non-linear insights
- Validation on external data and robustness testing
- Deployment of the GAM for real-time predictions

## Key Findings

- **GAM:** Best model ( $\text{Pseudo-R}^2 = 0.892$ ,  $\text{RMSE} = \$315 \text{ k}$ )
- **Ridge:** Strong linear baseline ( $\text{R}^2 = 0.858$ ,  $\text{RMSE} = \$370 \text{ k}$ )
- **Non-linearities:** Significant complex relationships present
- **Interpretability:** GAM retains clarity of feature effects

## Business Recommendation

Implement the GAM for price prediction, focusing on the non-linear relationships identified for range, ROC mean, and power per distance.

# Thank You for Your Attention

Questions?

*Statistical Learning Project – Aircraft Price Prediction*